

SIMATS ENGINEERING
Slot: C, Course Code: ITA0501, Course Name: Computer Vision

Duration: 60 mins

Activity Tool 2: Case Study (Medium)

Marks: 20

Assessment Tool Weightage: 20%

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Q. No.	Questions
1	Preprocessing and Feature Detection A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent. Analyze the issue and explain how preprocessing improves feature detection performance.
2	Hough Transform in Noisy Images A system uses Hough Transform to detect lines in noisy images but produces inaccurate results. Analyze the effect of noise on detection and suggest improvements based on preprocessing or parameter selection.
3	PCA for Feature Optimization A system extracts a large number of features, leading to high computational cost and redundancy. Analyze how Principal Component Analysis helps in reducing dimensionality and improving efficiency.
4	Corner Detection vs Edge Detection in Tracking A tracking system uses edge detection but fails to track moving objects accurately across frames. Analyze why edges are insufficient and justify the use of corner detection for reliable tracking.

Code of Conduct:

*I **SOMISETTY VENKATA SAI NAVEEN KUMAR (192425176)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student
SOMISETTY VENKATA SAI
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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Preprocessing and Feature Detection

Problem:

Images captured under different lighting conditions may have changes in brightness, contrast, shadows, and illumination. These variations can make feature detectors produce different or unreliable features.

Analysis:

Preprocessing improves the consistency and quality of feature detection by reducing unwanted variations in the image.

Useful preprocessing techniques:

- **Grayscale conversion:** Reduces computational complexity while retaining intensity information.
- **Noise removal:** Gaussian or median filtering removes unwanted noise.
- **Contrast enhancement:** Histogram equalization or CLAHE improves visibility of features.
- **Normalization:** Makes intensity values more consistent between images.
- **Illumination correction:** Reduces the effect of uneven lighting.

Result:

After preprocessing, edges and corners become clearer and more stable. This improves the accuracy and repeatability of methods such as Harris Corner, SIFT, ORB, and Canny Edge Detection.

2. Hough Transform in Noisy Images

Problem:

The Hough Transform detects geometric shapes by accumulating votes from image pixels. Noise creates unwanted edges, which also contribute votes to the Hough accumulator. As a result, false lines or inaccurate line parameters may be detected.

Analysis:

Noise can produce:

- False edge points
- Incorrect line detections
- Multiple unwanted lines
- Weak or fragmented actual lines

Improvements:

1. Apply **Gaussian or median filtering** before edge detection.
2. Use **Canny Edge Detection** to obtain cleaner edges.
3. Select suitable **Hough threshold values** to reject weak lines.
4. Adjust the **resolution of ρ (rho) and θ (theta)** according to the application.
5. Use a minimum line length and maximum gap when using **Probabilistic Hough Transform**.

Result:

Proper preprocessing combined with optimized Hough parameters reduces false detections and improves the accuracy of detected lines.

3. PCA for Feature Optimization

Problem:

A system extracting hundreds or thousands of features may contain redundant or highly correlated information. Processing all these features increases computational cost and may reduce model efficiency.

How PCA helps:

Principal Component Analysis (PCA) transforms the original features into a smaller set of new features called **principal components**.

The main steps are:

1. Standardize the feature data.
2. Calculate the covariance matrix.
3. Find eigenvalues and eigenvectors.

4. Rank components according to their explained variance.
5. Select the most important components.
6. Transform the original data into the reduced feature space.

For example, if a system has **100 features**, PCA may reduce them to **15–20 principal components** while retaining most of the important information.

Benefits:

- Reduces dimensionality
- Removes redundancy
- Decreases memory usage
- Reduces computational time
- Can improve model performance by reducing irrelevant information

Result:

PCA provides a compact representation of the data while preserving the most significant variations.

4. Corner Detection vs Edge Detection in Tracking

Problem:

Edge detection identifies boundaries where image intensity changes significantly. However, an edge is generally a line-like structure and does not provide a unique, stable location for tracking an object across frames.

Why edges are insufficient:

- Edges can be long and ambiguous.
- Object boundaries may change with movement.
- Edges can disappear because of lighting or occlusion.
- It is difficult to determine a unique correspondence between edges in consecutive frames.

Why corners are better:

Corners are points where intensity changes significantly in **two directions**. They provide distinctive and localized features that can be matched between frames.

Common corner detectors include:

- **Harris Corner Detector**
- **Shi-Tomasi Corner Detector**
- FAST

For example, when tracking a moving vehicle, corners around its windows, headlights, or other distinctive structures can be detected and matched from one frame to the next.

Result:

Corner detection provides stable, distinctive points for feature matching and motion estimation. Therefore, it is generally more suitable than edge detection for reliable object tracking across video frames.